

1. Administrative & Study Identification

Full Study Title: A Phase 1, Open-Label, Dose-Finding Study of BMS-986453, Dual Targeting BCMAxGPRC5D Chimeric Antigen Receptor T Cells, in Participants With Relapsed and/or Refractory Multiple Myeloma **Short Title/Acronym:** A Study to Assess BMS-986453 in RRMM **Protocol Number:** CA119-0002 / 24-051 **NCT Number:** NCT06153251 **Trial Status:** RECRUITING **Sponsor Name:** Juno Therapeutics, Inc. (A Bristol-Myers Squibb Company) / Bristol Myers Squibb **Investigational Product:** BMS-986453 (Dual Targeting BCMAxGPRC5D Chimeric Antigen Receptor T Cells) **Phase of Development:** Phase I **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead / Principal Investigator

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and to find the best dose of BMS-986453 in participants with relapsed and/or refractory multiple myeloma (RRMM). **Secondary Objectives:** To assess preliminary efficacy (e.g., objective response rate), duration of response, and the pharmacokinetics of the CAR T-cell therapy.

2.2 Background & Rationale To address the critical unmet medical need for patients with multiple myeloma whose disease has returned or continues to grow despite completing standard initial therapies. BMS-986453 utilizes cellular immunotherapy, modifying the patient's own T cells to dual-target BCMA and GPRC5D proteins on myeloma cells, thereby offering a novel approach to find and destroy treatment-resistant cancer cells.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Dose-Finding) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 150 study participants **Number of Sites:** Multicenter across the United States (including MSKCC, Stanford University, UCSF, City of Hope, UAB) **Estimated Study Start:** January 2024 **Estimated Primary Completion:** May 2030 (estimated)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participants must have a diagnosis of multiple myeloma with relapsed and/or refractory disease. 2. Confirmed progressive disease on or within 12 months (measured from the last dose) of completing treatment with the last anti-myeloma treatment regimen before study entry. 3. Participants in Part A and Part B Cohort 1 and in Part B Cohort 2 must have relapsed/refractory multiple myeloma and received previous antimyeloma therapy, including a proteasome inhibitor and an immunomodulatory agent, and have an Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. 4. Participants must have adequate organ function.

4.2 Exclusion Criteria 1. Any known active or history of central nervous system (CNS) involvement of multiple myeloma. 2. Active or history of plasma cell leukemia, Waldenstrom's macroglobulinemia, POEMS (polyneuropathy, organomegaly, endocrinopathy, monoclonal protein, skin changes) syndrome, or clinically significant amyloidosis. 3. History or presence of clinically significant CNS pathology such as seizure disorder, aphasia, stroke, severe brain injury, dementia, Parkinson's disease, or cerebellar disease, or presence of clinically active psychosis.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single dose of BMS-986453. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Administered as a single infusion following 3 days of lymphodepleting therapy. The lymphodepleting regimen includes fludarabine and cyclophosphamide (Cyclophosphamide ATC Code: L01AA01; Trade names: Cytosan, Endoxana).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care, infection prophylaxis, and management therapies for cellular therapy-related toxicities. **Prohibited:** Concurrent anti-myeloma treatments or use of other investigational therapeutic agents during the primary evaluation period.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to -1 Informed Consent, Medical History, Leukapheresis, Eligibility Labs
Baseline	Day 0	Lymphodepletion, BMS-986453 Infusion, Continuous Observation	Interim
Interim	Weeks 1-4	Safety Labs, Efficacy Assessment, Monitoring for CRS/ICANS	End of Study
End of Study	May 2030	Final Vital Signs, Exit Interview, Long-term Survival Follow-up	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: Incidence of dose-limiting toxicities (DLTs), adverse events (AEs), and determination of the maximum tolerated dose (MTD) / recommended Phase 2 dose, specifically measured by: 1. Number of participants with treatment-emergent adverse events (AEs). 2. Number of participants with serious adverse events (SAEs). 3. Number of participants with AEs leading to discontinuation. **Measurement Method:** Clinical observation and laboratory grading utilizing standard toxicity criteria (CTCAE).

7.2 Secondary & Exploratory Endpoints Objective Response Rate (ORR), Duration of Response (DoR), and progression-free survival. Pharmacokinetic parameters of dual-targeting CAR T cells in peripheral blood, specifically measured by: 1. Maximum observed concentration (C_{max}). 2. Time of maximum observed concentration (T_{max}). 3. Area under the blood concentration-time curve from time zero to 28 days after dosing (AUC(0-28D)).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Intensive monitoring via inpatient hospital care immediately post-infusion, followed by frequent outpatient checks. **Serious Adverse Events (SAEs):** Reporting timeline within 24 hours, with specific attention to Cytokine Release Syndrome (CRS) and neurotoxicity (ICANS). **Data Monitoring Committee (DMC):** Internal/External Dose Escalation Committee to review safety data before advancing to higher dose cohorts.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients who receive the lymphodepleting chemotherapy or the CAR T infusion) and Efficacy Evaluable Set. **Sample Size Justification:** N = 150, determined as statistically appropriate for a Phase 1 dose-escalation/expansion design to assess safety profiles and early signals of efficacy. **Handling of Missing Data:** Descriptive statistics for safety; right-censoring methods for time-to-event efficacy analyses (e.g., survival data).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite and remote monitoring due to the complexity and risks associated with Phase 1 cellular therapy. **Audit Trail:** Robust electronic Case Report Form (eCRF)

locking and data clarification mechanisms for strict regulatory adherence.

11. Study Identifiers & Governance

Posting ID: 8265227443882228 **Primary ID:** CA119-0002 **Registry Number:** NCT06153251 **Sponsor/Lead Organization:** Juno Therapeutics, Inc. / Bristol Myers Squibb **Study Title:** A Study to Assess BMS-986453 in Participants With Relapsed and/or Refractory Multiple Myeloma **Official Regulatory Title:** A Phase 1, Open-Label, Dose-Finding Study of BMS-986453, Dual Targeting BCMAxGPC5D Chimeric Antigen Receptor T Cells, in Participants with Relapsed and/or Refractory Multiple Myeloma

12. Clinical Framework (Study Specific)

Primary Condition: Relapsed and/or Refractory Multiple Myeloma (RRMM) **Diagnosis Threshold:** Confirmed progressive disease on or within 12 months of completing the last anti-myeloma treatment regimen **Medical Methodology:** Cellular Immunotherapy (Dual-Targeting CAR T Cells) **Optimization Target:** Identification of the optimal safe and efficacious dose

13. Study Procedures & Methodology

Management Pathway: Specialized CAR T Cell Therapy administration and post-infusion care protocol **Education Protocol (HCP):** Site-specific training on the preparation, handling, and administration of BMS-986453, plus rapid response protocols for serious side effects **Education Protocol (Patient):** Education on expected CAR T-cell side effects, requirements to stay near the treatment center, and caregiver support necessities **Quality Audit Mechanism:** Real-time pharmacovigilance tracking and dose-escalation review meetings

14. Observation & Follow-Up Schedule

Total Duration: Extends up to 5+ years for required gene therapy safety follow-up (Estimated completion May 2030) **Onsite Visit Cadence:** Intensive (daily/weekly) during the first month post-infusion, extending to quarterly, and eventually annual visits **Remote Monitoring Frequency:** Intermittent telehealth and remote checks as appropriate during the long-term follow-up phase **Checklist Review Interval:** Continuous AE/SAE reviews dynamically adjusted around dose administration

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator		[Insert Name]		_____		[DD-MMM-YYYY]										
	Clinical Operations Lead		[Insert Name]		_____		[DD-MMM-YYYY]										
		Lead Statistician		[Insert Name]		_____		[DD-MMM-YYYY]									